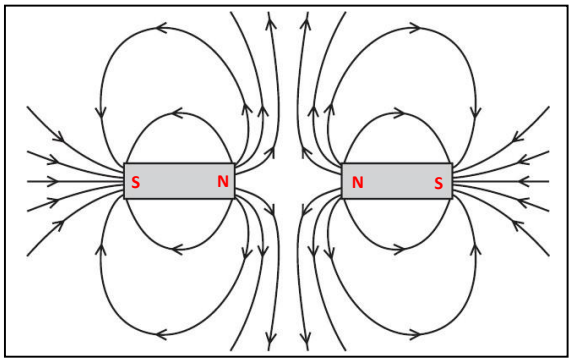
Magnetism

1. Law of magnetism:

   Two same pole will repel, while opposite pole will attract

2. **N** part of compass will always show direction of (geographical south pole/ geographical north pole) because it is attracted by (magnetic south pole/magnetic north pole).



3. What is showed by arrow in this picture?

   Magnetic field lines

4. What happened when iron material is placed in the area filled with arrow? It becomes magnet although it will be temporary as long as it is in the magnetic field of the permanent magnet

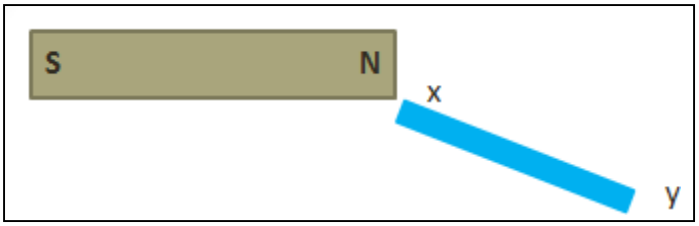
5. Which one is S and N pole from this picture?
   See the picture

6. It is known that there are 3 specimens (X, Y, and Z). But it is unknown which specimen is made of **wood**, **soft iron**, or **steel magnet**. To identify it, we test the end A and end B of the specimen with permanent magnet.

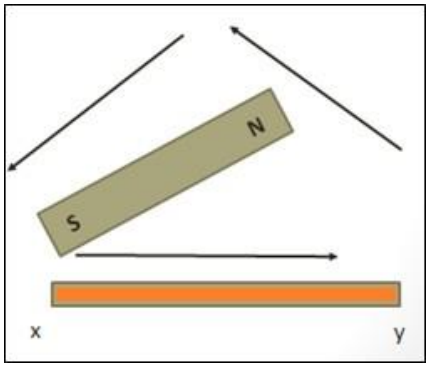
Below is the result of test. Fill the table with the correct material

Specimen	End A	End B	Material	Reason
X	No effect	No effect	Wood	Because wood doesn't cause any attraction or repulsion force
Y	Attracted	Repelled	Steel magnet	One side would be an attracted north pole and the other would be a repelled north pole because Steel is a hard magnetic material
Z	Attracted	Attracted	Soft Iron	because soft iron is a soft magnetic material so it can be lost of its magnetism easily and the poles are relative (easily changed)

7. The blue object is a magnetic material, when it is placed near magnet, it will become a magnet with x side become South pole, and Y side become North pole

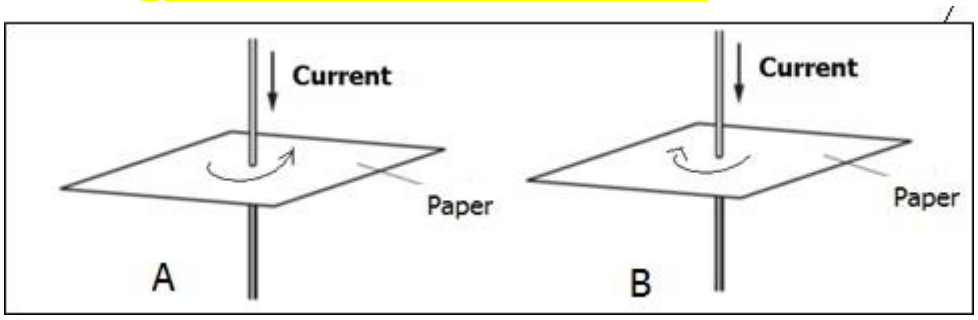


8. Orange rod is made of magnetic material. In this magnetization process, after several stroke **x** will become South pole, and **y** become North pole

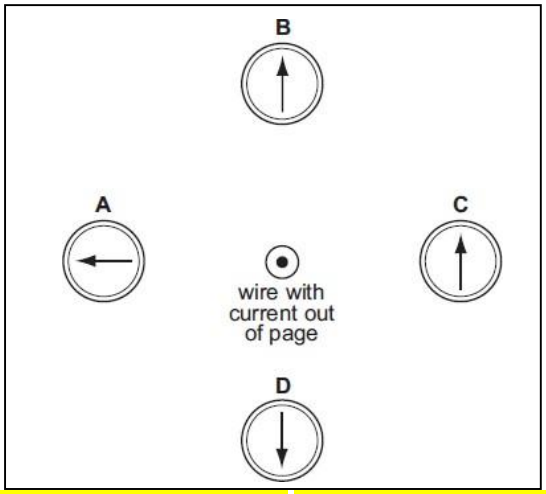


Electromagnet

1. Which is the correct drawing of magnetic field on the paper, because of electric current from the wire? **B and you can check it by using right hand-rule**



2. Which compass shows the correct direction of magnetic field generated by the wire? **C**

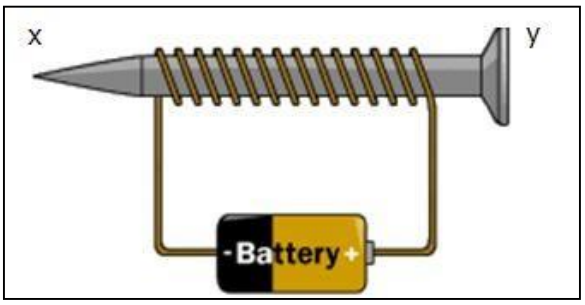


and you can check it by using right hand-rule with your thumb is out of the paper(page).

9. Electromagnet is magnet created by electricity. Why this kind of magnet is more useful?
   Because it is temporarily magnetic material. Once you cut off the electricity then its magnetism will lose too. You can activate its magnetism only by giving electric current and utilize it for certain purposes only. It will be more efficient and useful.

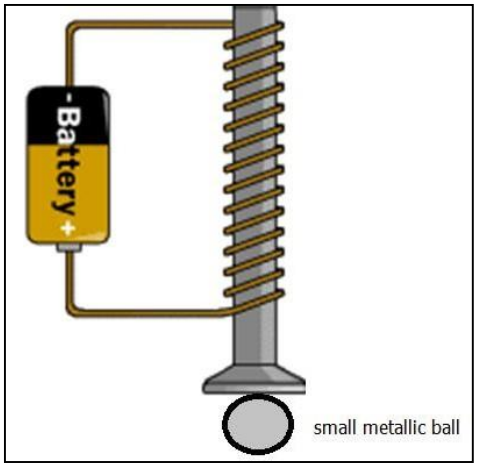
10. Give two examples of electromagnetic used in daily life!
    Circuit breakers
    Loudspeakers
    Computer hard disk drive
    Doorbells
    Headphones
    Incandescent light bulbs

11. In this process part x will become North pole, and y become South pole



12. Why electromagnets use soft iron type of material instead of steel?
    Because soft iron is a soft magnetic material that is easy to magnetize and lose its magnetism, we can say it is used to be a temporary magnet. These characteristics are helpful for specific purposes that need to control the change of attraction and movement of an object.

13. Suggest two ways to make electromagnet stronger!
    - Increase the magnitude of the electric current
    - Use more number coils of wire
    - Use high quality soft iron

14. The picture shows a small metallic ball is attracted to electromagnet (battery with iron nail). After several minutes, the battery is taken, but the small ball still attached. Explain why it happens!

    The magnetism is still in the iron nail a couple of times, and then the metal ball sticks to the nail. After a long time. It will fall because the iron nail's magnetism will lose soon.